

End Semester Regular and Supplementary Examination 2022-23

Digital Logic (CSEUGPC02)

B Tech , 2nd Year, 3rd Sem

FM: 80

Time : 3.00 Hrs

(Answer 10 marks from Section A, 30 marks from Section B, 40 marks from Section C)
Symbols have their usual meaning. Answer question 1 in one place.

Section A

1. Answer any five Questions (5x2=10)

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|---|---|
| i. Implement $f = (AB + CD)$ using switches only | vi. Convert $(52)_8$ to Hexadecimal number A2 |
| ii. Complement the function $f = X \cdot (Y'Z' + YZ)$ | vii. Converts Gray [1101] to Binary number |
| iii. What is Operator Precedence in Boolean Algebra | viii. Convert $(01010101)_2$ to Decimal number |
| iv. What is relation between Minterm and Maxterms. | ix. Prove, $A + A'B = A + B$ |
| v. Represent (-19) in 2's complement form | x. Compare frequency of N-bit Ring and Johnson counter. |

Section B

Answer any six Questions (6x5=30)

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|---|---|
| 2. Design a 2:1 Multiplexer circuit | 7. Prove, $(XY + X'Z) = (X + Z)(X' + Y)$ |
| 3. Describe a Enabled RS Latch | 8. Prove, $BC + AC' + AB = BC + AC'$ |
| 4. Minimize $(A+BC)(A+B') + A'B'$ | 9. Reduce the function, $A[B+C'(AB+AC)']$ |
| 5. Find the sum of minterms of $F(A,B,C) = A+B'C$ | 10. Perform addition $(84 + 76)$ in BCD form. |
| 6. State and prove a D'Morgan Theorem | 11. Design a Half adder using five NAND Gates only. |

Section C

Answer any four Questions (4x10=40)

12. Convert $(41.6875)_{10}$ to binary number, then map to Octal and Hexadecimal number. (6+2+2)
13. Perform addition $(-32) + (40)$ using signed 2's complement method. Now perform the subtraction $(32) - (40)$ using signed 2's complement method. (5+5)
14. $F(A,B,C,D) = \sum_m (0,2,3,5,7,8,9,10,11,13,15)$. Solve the essential prime implicants and the Boolean functions (5+5)
15. $F(A,B,C,D) = \sum_m (0,1,2,5,8,9,10)$. Implement the SOP and POS function using basic gates (5+5)
16. Implement a Full Adder circuit using two 4:1 Multiplexers (10)
17. What is Full Subtractor circuit? Design a Full Subtractor using a 3:8 Decoder Circuit (6+4)
18. Draw a state diagram and state table of a 2 bit up-counter that holds state when input $x=0$, otherwise counts up in sequence. Design the said counter using DFF. (4+4+2)